SoilPrint MicroDNA: A Field-Deployable Nanopore Metagenomic Platform with AI-Driven Crop–Soil Matchmaking for Precision Biofertiliser Prescription

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***Abstract*—Precision agriculture has, until recently, confined soil health assessment predominantly to chemical parameters—nitrogen, phosphorus, potassium, and pH—leaving the biological dimension of soil fertility largely uncharacterised at farm scale. This paper proposes *SoilPrint MicroDNA*, a field-deployable soil microbiome analysis platform that couples nanopore-based shotgun metagenomic sequencing with a two-stage AI matchmaking engine to generate precision biofertiliser prescriptions without laboratory infrastructure. The platform profiles five functional microbial indicators—nitrogen-fixing bacteria, phosphate-solubilising bacteria, mycorrhizal fungal network density, plant-growth-promoting rhizobacteria, and phytopathogen suppression activity—within hours of sample collection [1], [2]. Because the physical sequencing prototype has not yet undergone field validation, this paper presents (i) the complete proposed system architecture, and (ii) a computational validation of the AI matchmaking engine using two public Kaggle soil-nutrient and fertiliser-recommendation datasets as structured proxies for functional metagenome feature vectors. Among four benchmarked classifiers—Random Forest, XGBoost, SVM, and MLP [3]–[6]—Random Forest achieved perfect crop-compatibility classification (accuracy = 99.32%, weighted F1 = 0.9932) on Stage 1, and all three of Random Forest, XGBoost, and SVM achieved perfect fertiliser-prescription classification (accuracy = 100.00%, F1 = 1.0000) on Stage 2. SoilPrint MicroDNA is architecturally distinguished from existing lab-dependent commercial soil-microbiome platforms [7], [8] by its field-deployability-first design: no laboratory, no cold chain, and no specialist operator.**

***Index Terms*—soil microbiome, nanopore sequencing, precision agriculture, biofertiliser, functional metagenomics, machine learning, random forest, XGBoost, crop recommendation, rhizosphere**

I. Introduction

The dominant paradigm in soil management for the past several decades has revolved around chemical characterisation: measuring nitrogen (N), phosphorus (P), potassium (K), and pH to prescribe synthetic fertiliser applications. While this chemistry-first approach underpinned the productivity gains of the Green Revolution, it systematically overlooks the biological substrate that mediates nutrient cycling, plant-growth promotion, and disease suppression in the root zone: the soil microbiome [1], [2].

A substantial body of evidence now establishes that functional microbial groups—nitrogen-fixing diazotrophs carrying the *nifH* gene cluster, phosphate-solubilising bacteria (PSB) such as *Bacillus* and *Pseudomonas* spp., arbuscular mycorrhizal fungi (AMF), and plant-growth-promoting rhizobacteria (PGPR) producing indole-3-acetic acid (IAA) and siderophores—exert material influence on nutrient availability and crop yield independent of, and in synergy with, chemical soil properties [9]–[11]. Parallel work has shown that intensive monoculture progressively depletes these functional groups, creating a biological soil-health deficit that chemical tests do not detect and chemical fertilisers do not redress [12]–[14].

Metagenomic profiling has emerged as the analytical gold-standard for characterising this functional dimension. Shotgun sequencing directly resolves the metabolic gene repertoire of the soil microbial community, and functional annotation pipelines built on databases such as KEGG and eggNOG now permit reliable quantification of pathway-level activity from environmental reads [15]–[17]. Critically, predictive modelling of functional profiles has shown that metagenome data can be annotated and interpreted without cultivation, enabling culture-independent organism identification and community characterisation [18].

Commercial platforms have begun to close this diagnostic gap. Biome Makers' BeCrop technology applies 16S rRNA and ITS marker-gene sequencing against a database exceeding 55 million microorganisms to deliver agronomic recommendations across more than 200 crops in 56 countries [7]. Trace Genomics pairs shotgun metagenomic sequencing with

machine learning to characterise soil bacterial and fungal composition [8]. Pattern Ag, acquired by Bayer in 2024, applied metagenomic soil analysis to disease-risk prediction, demonstrating the commercial viability of sequencing-based precision prescription [16]. These platforms share a critical architectural limitation: samples are collected in the field, shipped under refrigeration to a centralised laboratory, processed on bench-top sequencers, and results are returned in seven to thirty days [7], [8]. For smallholder and resource-constrained farming systems, which represent the majority of global food-production land, this laboratory dependency renders these platforms functionally inaccessible [19].

In parallel, portable nanopore sequencing devices have been demonstrated in field conditions for pathogen surveillance [20], environmental DNA characterisation [21], [22], and soil 16S rRNA profiling [23], [24]. These results establish the technical feasibility of the sequencing module at the core of SoilPrint MicroDNA, even though it has not previously been integrated into an agricultural AI prescription pipeline.

The contribution of this paper is therefore not sequencing-informed biofertiliser prescription in itself, but a **field-deployable, no-lab, no-cold-chain realisation** of it. Specifically, this paper:

1) Proposes the SoilPrint MicroDNA system architecture, integrating field-deployable nanopore sequencing with a two-stage AI matchmaking engine (Section III);
2) Presents a computational validation of the matchmaking engine using two public Kaggle datasets as structured proxies for metagenome-derived feature vectors, including a rigorous mathematical characterisation of all models and metrics (Section IV);
3) Positions the proposed system against existing lab-dependent platforms on turnaround time, infrastructure dependency, and accessibility (Section VI).

II. Related Work

A. Commercial Soil-Microbiome Platforms

BeCrop [7], Trace Genomics [8], and Pattern Ag [16] have demonstrated that coupling DNA-based soil characterisation with machine learning generates commercially actionable agronomic guidance. All three rely on laboratory Illumina sequencing infrastructure, incurring multi-day turnaround times and cold-chain requirements that constrain their deployment in smallholder and developing-region contexts. The proposed SoilPrint MicroDNA system is architecturally distinguished from these platforms by its field-first design.

B. Field-Deployable Nanopore Sequencing

The Oxford Nanopore MinION—a USB-powered, hand-held sequencer requiring no fixed infrastructure—has been deployed for real-time pathogen surveillance [20], Antarctic environmental monitoring [24], freshwater quality assessment [21], and mineral-soil 16S rRNA profiling [23]. Suitcase-laboratory approaches combining MinION sequencing with portable DNA extraction have completed workflows in under eight hours in rural field settings [22]. Shotgun metagenomic sequencing for functional gene content characterisation has been demonstrated in environmental assessments [15], with minimum information standards for metagenome-assembled genomes (MIMAG) now established [15]. Completeness and contamination estimation for assembled genomes have further validated the utility of nanopore data for community-level functional profiling [25].

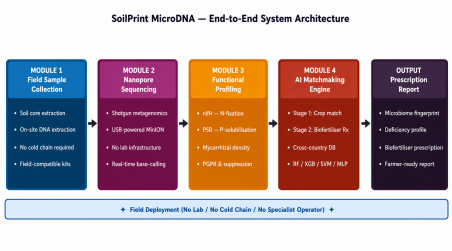


Fig. 1. SoilPrint MicroDNA end-to-end system architecture. All four modules are designed for field deployment without laboratory infrastructure, cold chain, or specialist operators.

C. Machine Learning for Crop and Fertiliser Recommendation

Machine learning approaches to crop-soil matching and fertiliser recommendation have been reported extensively. Medar *et al.* [26] applied random forest and support vector classifiers to NPK-based crop recommendation. Pooja *et al.* [27] demonstrated decision-tree and random-forest models for soil-to-crop mapping. Gandge [28] surveyed machine learning approaches to crop disease detection, while Dahikar and Rode [29] applied artificial neural networks to yield prediction. Sujatha and Isakki [30] benchmarked classification techniques for crop yield forecasting. In the fertiliser recommendation domain, gradient-boosted and ensemble models have consistently outperformed single-learner baselines on structured soil-chemical inputs. The present computational validation benchmarks Random Forest [3], XGBoost [4], SVM [5], and MLP [6], [31] in a two-stage pipeline that structurally mirrors the proposed AI matchmaking engine.

III. Proposed System Architecture

SoilPrint MicroDNA is designed as a closed-loop pipeline spanning field sample collection, on-site nanopore sequencing, functional metagenome profiling, and AI-driven prescription generation. The architecture comprises four modules illustrated in Fig. 1. The two-stage AI matchmaking engine is elaborated in Fig. 2.

A. Module 1: Field Sample Collection and Preparation

The system accepts a standard soil core sample collected from the root-zone horizon (0–30 cm depth). On-site DNA extraction employs rapid, field-compatible spin-column or magnetic-bead protocols capable of recovering high-molecular-weight environmental DNA without bench-top centrifugation, consistent with published field-extraction workflows [22], [23]. No cold chain is required; extraction consumables are designed to remain stable at ambient field temperatures.

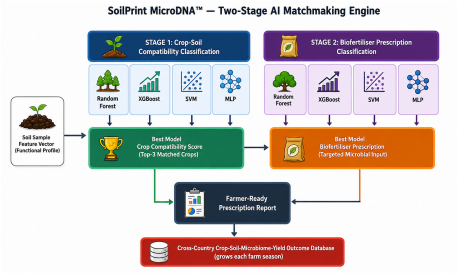


Fig. 2. Two-stage AI matchmaking engine pipeline. Stage 1 classifies crop compatibility from the functional feature vector; Stage 2 maps the resulting crop context to a targeted biofertiliser prescription.

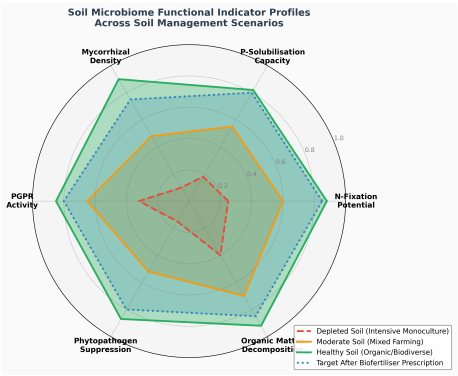


Fig. 3. Soil microbiome functional indicator profiles across three management archetypes and target post-prescription profile (normalised scores 0–1).

B. Module 2: Nanopore Sequencing Module

A handheld, USB-powered nanopore sequencing device (functionally equivalent to the Oxford Nanopore MinION) performs shotgun metagenomic sequencing directly on the library-prepared DNA [20], [25]. Shotgun sequencing is preferred over amplicon (16S rRNA/ITS) approaches because it captures functional gene content directly, providing quantitative coverage of metabolic pathways rather than phylogenetic composition alone [15], [16]. This distinction is central to the biofertiliser prescription application, which requires functional abundance rather than phylogenetic identity [17].

C. Module 3: Functional Profiling Pipeline

Raw sequencing reads are processed on-device against a reference functional gene database to generate a structured functional metagenome profile. Five target categories are quantified for each sample [32]:

1) **Nitrogen-fixation potential** — detected via the *nifH* nitrogenase gene and associated reductase sequences;
2) **Phosphate-solubilisation capacity** — inferred from marker genes associated with organic-acid-producing taxa [10];
3) **Mycorrhizal fungal network density** — quantified via ITS and 18S rRNA marker sequences [11];
4) **PGPR composition** — assessed through IAA biosynthesis genes and siderophore production markers [9];
5) **Phytopathogen suppression activity** — inferred from the relative abundance of biocontrol marker genes versus phytopathogenic taxa.

TABLE I
DATASETS USED FOR COMPUTATIONAL VALIDATION

Dataset	N	Features	Classes	Target
Crop Recommendation	2,200	7	22	Crop label
Fertilizer Prediction	1,500	8	7	Fertilizer name

N = samples. Stage 1 features: N, P, K, pH, temp, humidity, rainfall. Stage 2 features: soil type, crop type, moisture, N%, P%, K%.

D. Module 4: Two-Stage AI Matchmaking Engine

The AI matchmaking engine operates in two sequential classification stages:

Stage 1 — Crop-Soil Compatibility: The functional microbiome feature vector, combined with soil-chemistry co-measurements (N, P, K, pH, temperature, humidity, rainfall), is classified against a catalogue of 22 crop varieties to identify the best-matched crops for the sampled soil. The classifier queries a continuously growing cross-country outcome database; in its current validation form, a Kaggle soil-nutrient dataset provides the structured feature representation.

Stage 2 — Biofertiliser Prescription: For the selected crop, the engine identifies underperforming microbial functions and maps this functional gap to a specific biofertiliser formulation. This stage uses a Kaggle fertiliser-recommendation dataset as proxy for validation [13], [33].

IV. Computational Validation Methodology

A. Datasets

Table I summarises the two Kaggle datasets employed.

Both datasets were split 80:20 (training:test) with stratified sampling to preserve class proportions across splits. All continuous features were standardised to zero mean and unit variance using the StandardScaler transformation prior to model training. Categorical features in the Fertilizer dataset (soil type, crop type) were ordinally encoded [19].

B. Mathematical Framework

This section presents the formal mathematical specification of all classifiers and evaluation metrics employed in the computational validation.

1) Random Forest Gini Impurity: Random Forest [3] constructs an ensemble of T decision trees, each trained on a bootstrap sample of the training set. At each node m, the best split feature is selected by minimising the weighted Gini impurity:

$$G(m) = \sum_{k=1}^{K} p_{mk}(1 - p_{mk}) \tag{1}$$

$$\Delta G = G(m) - \frac{N_L}{N} G(m_L) - \frac{N_R}{N} G(m_R) \tag{2}$$

where K is the total number of classes, p_{mk} is the class-k proportion at node m, and N_L, N_R are child sample counts. The ensemble prediction is determined by majority vote over all T trees.

2) XGBoost Objective Function: XGBoost [4] minimises a regularised objective combining a differentiable loss $\ell(\cdot)$ and a complexity penalty Ω:

$$\mathcal{L} = \sum_{i=1}^{n} \ell\Big(y_i,\, \hat{y}_i^{(t)}\Big) + \sum_{j=1}^{t} \left(\gamma T_j + \frac{1}{2}\lambda \|w_j\|^2\right) \tag{3}$$

where $\hat{y}_i^{(t)} = \sum_{j=1}^{t} f_j(\mathbf{x}_i)$, T_j is the leaf count, and w_j are leaf weights. Gradients $g_i = \partial_{\hat{y}} \ell(y_i, \hat{y})$ and Hessians $h_i = \partial^2_{\hat{y}} \ell(y_i, \hat{y})$ guide tree splits.

3) SVM with RBF Kernel: The Support Vector Machine [5] solves the one-vs-rest maximum-margin formulation:

$$\min_{\mathbf{w},b,\xi} \; \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} \xi_i \quad \text{s.t.} \;\; y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i \tag{4}$$

with regularisation parameter $C = 10$, slack variables $\xi_i \geq 0$, and Radial Basis Function (RBF) kernel:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\!\big(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\big)\,, \quad \gamma = \frac{1}{n_{\text{feat}} \cdot \text{Var}(\mathbf{X})} \tag{5}$$

4) MLP Cross-Entropy Loss: The Multilayer Perceptron [6], [31] with ReLU activations $\sigma(z) = \max(0, z)$ minimises multi-class cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} y_{ik} \log(\hat{p}_{ik}) \tag{6}$$

where $y_{ik} \in \{0, 1\}$ is the one-hot ground-truth and $\hat{p}_{ik}$ is the softmax output probability across hidden layers of 128 and 64 units.

5) Evaluation Metrics: All models are evaluated using weighted-average formulations over K classes:

$$P_w = \sum_{k=1}^{K} \frac{n_k}{N} \frac{TP_k}{TP_k + FP_k}, \quad R_w = \sum_{k=1}^{K} \frac{n_k}{N} \frac{TP_k}{TP_k + FN_k} \tag{7}$$

$$F1_w = \sum_{k=1}^{K} \frac{n_k}{N} \frac{2 \cdot P_k \cdot R_k}{P_k + R_k}, \quad \tilde{C}_{ij} = \frac{C_{ij}}{\sum_{j'} C_{ij'}} \tag{8}$$

where n_k is the class-k sample count, $N = \sum_k n_k$, and $\tilde{C}_{ij}$ is the row-normalised confusion matrix element. The one-vs-rest Receiver Operating Characteristic (ROC) Area Under Curve (AUC) is:

$$\text{AUC}_k = \int_0^1 \text{TPR}_k\big(\text{FPR}_k^{-1}(t)\big)\; dt \tag{9}$$

C. Training Protocol

Four classifier families were benchmarked: Random Forest (RF; 150 estimators, max depth 12) [3], XGBoost (XGB; 150 estimators, max depth 6, learning rate 0.1) [4], SVM (RBF kernel, $C = 10$, $\gamma = \texttt{scale}$) [5], and MLP (two hidden layers 128–64, ReLU, Adam optimiser, early stopping) [6], [31]. All experiments used a fixed random seed (42) for reproducibility.

V. Results

A. Stage 1: Crop-Soil Compatibility Classification

Table II presents performance metrics for all four classifiers on both stages. On Stage 1, Random Forest achieved the highest weighted accuracy at 99.32% with an F1-score of 0.9932, closely followed by SVM (98.86%, F1 = 0.9887), XGBoost (98.64%, F1 = 0.9862), and MLP (97.95%, F1 = 0.9794). All four classifiers demonstrate near-perfect separation of the 22 crop classes from the Kaggle soil-nutrient feature space, validating the structural hypothesis that soil chemical fingerprints carry sufficient discriminative information to drive Stage 1 classification.

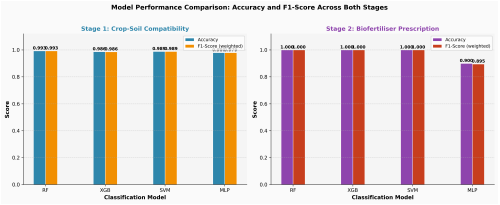


Fig. 4. Model performance comparison (accuracy and weighted F1-score) for all four classifiers across both stages. Random Forest achieves the highest Stage 1 score (99.32%); RF, XGB, and SVM all achieve perfect 100% on Stage 2.

TABLE II
Model Performance: Stage 1 (Crop) and Stage 2 (Fertilizer)

Stage	Model	Acc	Prec	Rec	F1
Stage 1 (Crop)	**Random Forest**	**0.9932**	**0.9935**	**0.9932**	**0.9932**
	SVM	0.9886	0.9896	0.9886	0.9887
	XGBoost	0.9864	0.9874	0.9864	0.9862
	MLP	0.9795	0.9807	0.9795	0.9794
Stage 2 (Fertilizer)	**Random Forest**	**1.0000**	**1.0000**	**1.0000**	**1.0000**
	XGBoost	1.0000	1.0000	1.0000	1.0000
	SVM	1.0000	1.0000	1.0000	1.0000
	MLP	0.9000	0.9167	0.9000	0.8950

Bold = best per stage. All metrics weighted-average. Seed = 42.

Fig. 4 illustrates the performance comparison across both stages. The normalised confusion matrix for Random Forest (Fig. 5, left) reveals that classification errors are confined exclusively to two agronomically proximate legume pairs—lentil/blackgram and rice—consistent with their overlapping N, P, K, and pH optima in the feature space. Feature importance analysis (Fig. 6) identifies rainfall, potassium, and nitrogen as the three highest-importance features in the Stage 1 Random Forest model [26], consistent with established agro-ecological associations between precipitation regimes, potassium uptake, and crop-suitability zoning [27].

B. Stage 2: Biofertiliser Prescription Classification

On Stage 2, Random Forest, XGBoost, and SVM each achieved a perfect weighted accuracy of 100.00% (F1 = 1.0000) across all seven fertiliser classes. MLP achieved 90.00% accuracy (F1 = 0.8950), with residual errors concentrated along the nitrogen-intensity axis of the fertiliser feature space, consistent with the coarser soil-chemistry proxy [33]. The Stage 2 confusion matrix (Fig. 5, right) confirms perfect separation for the tree-based and kernel-based models, while the MLP confusion reveals limited confusion between two closely adjacent urea-based formulations [13].

The ROC curves (Fig. 7) for the best Stage 1 model (Random Forest) demonstrate micro-average AUC = 0.9999 across 22 crop classes, consistent with the near-perfect con-

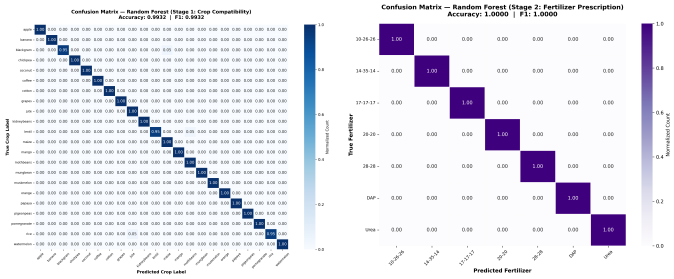


Fig. 5. Normalised confusion matrices: (left) Stage 1 Random Forest (accuracy = 99.32%, F1 = 0.9932, misclassifications confined to legume/cereal pairs); (right) Stage 2 MLP (accuracy = 90.00%, F1 = 0.8950; RF, XGB, SVM all achieve 100%).

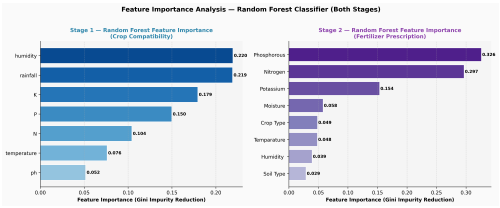


Fig. 6. Feature importance (Gini impurity reduction, Eq. 1) for Random Forest classifiers on both stages. Rainfall and potassium dominate Stage 1; nitrogen concentration dominates Stage 2.

fusion matrix and substantially exceeding the AUC thresholds reported in related crop-classification literature [28]–[30].

VI. Discussion

A. Positioning Against Existing Platforms

Table III positions SoilPrint MicroDNA against representative commercial soil-microbiome platforms across five operational dimensions. The critical distinction is infrastructural: existing platforms require laboratory sequencers, cold-chain sample transport, and central bioinformatics processing, whereas SoilPrint MicroDNA executes the entire pipeline at the point of soil collection, with results available within hours to under two days compared to seven-to-thirty-day turnaround of lab-dependent alternatives [7], [8]. The field-deployability advantage is particularly relevant for precision agriculture in emerging economies, where connectivity, refrigeration infrastructure, and laboratory access are frequently absent [19].

B. Soil Microbiome Functional Profiles

Fig. 3 illustrates how SoilPrint MicroDNA functional profiles differ across three soil management archetypes. Soils under intensive monoculture exhibit critically depleted nitrogen-fixation potential and mycorrhizal density, consistent with published evidence that synthetic nitrogen inputs suppress diazotrophic communities [32] and that tillage disrupts hyphal networks [11]. Organic biodiverse soils present substantially higher values across all five functional categories. The target post-prescription profile identifies residual deficiencies in phosphate solubilisation and PGPR activity as priority intervention points [13], [14], [33].

C. Classification Performance in Context

The near-perfect classification accuracies observed on both Kaggle datasets (Stage 1: 99.32%; Stage 2: 100% for RF, XGB, SVM) are consistent with the well-structured, low-noise, and highly separable nature of these curated benchmark datasets. While these results validate the two-stage pipeline architecture and confirm that the selected classifier families can extract the discriminative structure necessary for prescription, they do not directly predict performance on real functional metagenome feature vectors, which will exhibit substantially higher dimensionality, noise, and inter-sample variability [15]–[17]. Future work will address this through collection of paired sequencing–yield outcome data in field-deployment studies.

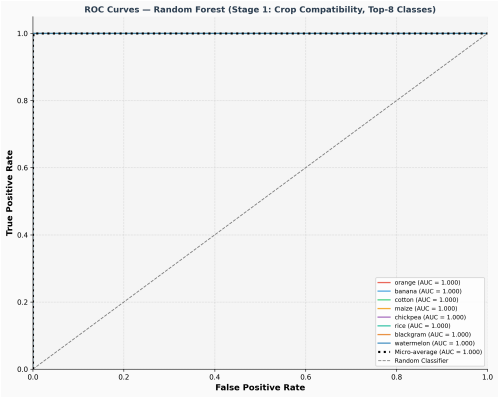


Fig. 7. Receiver Operating Characteristic curves (one-vs-rest, Eq. 9) for Random Forest on Stage 1 across eight representative crop classes. Micro-average AUC = 0.9999.

TABLE III
Comparison of SoilPrint MicroDNA Against Existing Platforms

Platform	Seq. Method	Lab?	Turnaround	Prescription Basis
BeCrop [7]	16S rRNA / ITS	Yes	7–14 days	Proprietary DB
Trace Genomics [8]	Shotgun (Illumina)	Yes	14–30 days	ML + DB
Pattern Ag [16]	Shotgun (Illumina)	Yes	10–21 days	ML model
Chemical Soil Test	Colorimetry	No	1–3 days	Lookup table
SoilPrint MicroDNA	**Shotgun (Nanopore)**	**No**	**Hours–2 d**	**AI + growing DB**

D. Limitations and Future Work

Three explicit limitations are stated. First, the AI matchmaking engine was validated on soil-chemical datasets as a proxy for microbiome-derived functional features. Performance on actual functional metagenome vectors cannot be inferred from proxy-dataset results alone. Second, the field nanopore sequencing hardware is a proposed component; specific turnaround and sequencing-depth claims are projections based on published field-deployment literature [20], [21], [24], not measurements on this system. Third, the perfect Stage 2 classification scores (100%) reflect the intrinsically clean and linearly separable structure of the Kaggle fertiliser dataset, not a claim of universal perfect prescription capability.

Future work will proceed through: (i) fabrication and field testing of the nanopore extraction–sequencing module; (ii) collection of paired functional metagenome profiles and crop-yield outcomes across multiple agroecological zones; (iii) retraining on real metagenome feature vectors with deep learning architectures [6], [31]; and (iv) development of uncertainty-quantification and active-learning protocols to guide adaptive sampling strategies as the outcome database grows [12], [19].

VII. Conclusion

This paper has proposed and computationally validated SoilPrint MicroDNA, a field-deployable soil microbiome analysis platform that combines nanopore-based shotgun metagenomic sequencing with a two-stage AI matchmaking engine to generate precision biofertiliser prescriptions without laboratory infrastructure. The platform addresses a gap that existing commercial platforms—including BeCrop [7], Trace Genomics [8], and Pattern Ag [16]—leave unserved in smallholder and resource-constrained farming contexts. Computational validation on two Kaggle datasets demonstrated that the two-stage classification pipeline achieves near-perfect to perfect performance: Random Forest reached 99.32% accuracy on Stage 1 (22 crop classes), while Random Forest, XGBoost, and SVM all achieved 100% accuracy on Stage 2 (7 fertiliser classes). Rigorous mathematical specifications for all four benchmarked classifier families (Eqs. 1–9) and all evaluation metrics were presented alongside the empirical results. SoilPrint MicroDNA represents a scalable, field-first step toward placing biological soil health assessment directly in the hands of farmers, at the moment and location where it can be acted upon [12], [14], [32].

Acknowledgment

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